Multi-view Interaction Reliability Filter (MIRF) Architecture

Axiomatic Synthesis of Statistical Consistency, LLM Review Auditing, and Temporal Burst Signals

Branch 1: Statistical Consistency (RRFN)

- Anchor Subsets: $\mathcal{A}_1, \ldots, \mathcal{A}_5 \subset \mathcal{I}$
- Transition Matrix: $T \in \mathbb{R}^{5 \times 5}$
- Tikhonov Inversion: $T_\lambda^\dagger = (T^T T + \lambda I)^{-1} T^T$
- Prediction Divergence: $|\tilde{r}_{ui} - \mathbb{E}_{\mathbf{p}}[Y^*]|$

$$\mathbf{R_{RRFN}(u, i)} = \exp\left(-\frac{|\tilde{\mathbf{r}}_{\mathbf{ui}} - \mathbb{E}[\mathbf{Y}^*]|^2}{2\sigma^2_{\mathbf{RRFN}}}\right)$$

Branch 2: Semantic Review Audit (LLM)

- Review Text Ingestion: $S_{ui} \in \mathcal{S}$
- Sentiment & Plausibility Verification
- Repetition & Shilling Bot Detection
- Thread-Safe SQLite WAL Cache

$$\mathbf{R_{LLM}(u, i) = LLM_Audit(S_{ui}, \tilde{r}_{ui}) \in [0, 1]}$$

Branch 3: Burst Detection (2D BLAS)

- Rolling Window Accel: $Z_{\text{temp}}(i, t)$
- Extreme Rating Skew: $S_{\text{pol}} = \frac{|N_1 - N_5|}{N_1 + N_5 + \varepsilon}$
- 2D BLAS Matrix Broadcasting
- Compute Time: < 25ms over 42K edges

$$\mathbf{R_{bomb}(u, i) = 1 - \sigma(\omega_1 Z_{temp} + \omega_2 S_{pol} - \theta)}$$

MIRF Convex Fusion & Reliability Weight Synthesis

$$\mathbf{w_{ui} = R_{MIRF}(u, i) = \alpha R_{RRFN}(u, i) + \beta R_{LLM}(u, i) + \gamma R_{bomb}(u, i)}, \quad \alpha + \beta + \gamma = 1$$

Default Defensive Calibration: $\alpha = 0.40,\ \beta = 0.30,\ \gamma = 0.30$ $\longrightarrow$ **Continuous Sample Reliability** $w_{ui} \in [0, 1]$

Consumer 1: Graph Contrastive Learning

Modulates InfoNCE Loss $\mathcal{L}_{\text{cl}} = -\sum w_i \log(\cdot)$ **and Risk Loss** $\mathcal{L}_{\text{risk}}$

Consumer 2: SUPER Blueprint Denoising

Filters Poisoned Edges: $B_u = \{i \mid w_{ui} \geq \theta_{\text{trust}}\}$ $\longrightarrow$ **Pure Tail Inclination** α_u